DRESS: A Continuous Framework for Structural Graph Refinement

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Abstract

The Weisfeiler-Lehman (WL) hierarchy is a cornerstone framework for graph isomorphism testing and structural analysis. However, scaling beyond 1-WL to 3-WL and higher requires tensor-based operations that scale as $\mathcal{O}(n^3)$ or $\mathcal{O}(n^4)$, making them computationally prohibitive for large graphs. In this paper, we present Generalized-DRESS, an abstract continuous framework parameterised by the choice of neighbourhood operator, aggregation function and norm. We introduce Motif-DRESS, a specialisation that defines neighbourhoods via structural motifs, and show that under three sufficient conditions it converges to a unique fixed point. The Original-DRESS equation (Castrillo, León, and Gómez, 2018) is recovered as a special case with triangle motifs. We then introduce Δ-DRESS, which runs standard DRESS on each node-deleted subgraph $G \setminus \{v\}$, connecting DRESS to the Kelly–Ulam reconstruction conjecture. We demonstrate empirically that both Motif-DRESS and Δ-DRESS distinguish Strongly Regular Graphs (SRGs)—such as the Rook and Shrikhande graphs—that confound standard 3-WL. Our results establish Generalized-DRESS as a highly scalable framework that empirically surpasses the distinguishing power of 3-WL on well-known benchmark graphs, without the prohibitive $\mathcal{O}(n^4)$ computational cost.

1 Introduction

The graph isomorphism problem and structural similarity measurement rely heavily on the Weisfeiler-Lehman (WL) hierarchy [7]. While the standard 1-WL test—shown to characterize the expressiveness of message-passing neural networks [8, 6]—is highly efficient, it fails to distinguish many non-isomorphic graphs. Higher-order k-WL tests [2] offer greater expressiveness, but their $\mathcal{O}(n^3)$ or $\mathcal{O}(n^4)$ complexity renders them impractical for large-scale network analysis.

The Original-DRESS algorithm [3] provides a continuous alternative: a parameter-free, non-linear dynamical system that iteratively refines edge weights based on local structural neighbourhoods. Original-DRESS operates over triangles (common neighbours) and is equivalent to 1-WL on the line graph $L(G)$. However, like 1-WL, it fails to distinguish highly symmetric structures such as Strongly Regular Graphs (SRGs).

In this work, we present a hierarchy of continuous structural refinement algorithms built on a common template:

1. **Generalized-DRESS:** An abstract framework parameterised by a symmetric neighbourhood operator, aggregation function and norm.

2. **Motif-DRESS:** A specialisation of Generalized-DRESS that defines neighbourhoods via structural motifs such as 4-cycles or K_4 cliques. Original-DRESS is recovered as a special case with triangle motifs.

3. **Δ-DRESS:** Runs standard DRESS on each node-deleted subgraph $G \setminus \{v\}$, connecting DRESS to the reconstruction conjecture and empirically distinguishing SRGs that confound 3-WL.

We demonstrate empirically that Motif-DRESS and Δ-DRESS distinguish graphs beyond the reach of 3-WL, while maintaining near-linear scalability.

Isomorphism Test. Throughout this paper, the **graph fingerprint** is the sorted vector of converged edge values $\mathrm{sort}(d^*)$. Two graphs are declared non-isomorphic if and only if their fingerprints differ element-wise beyond a numerical tolerance ϵ. This applies uniformly to all DRESS variants.

All experiments use a convergence tolerance of $\epsilon = 10^{-6}$ and a maximum of 20 iterations; in every case convergence was reached before the iteration limit.

Related Work. Higher-order GNN architectures that exceed 1-WL expressiveness include k-GNN [6], GIN [8], PPGN [5], and Ring-GNN [4]. Subgraph-based approaches such as GNN-AK+ [9] and ESAN [1] achieve higher expressiveness by running message-passing networks on node-deleted or node-marked subgraphs; Δ-DRESS (Section 4) shares the node-deletion strategy but replaces learned message passing with deterministic fixed-point iteration. All of the above are **supervised** methods that require labelled training data and learnable parameters. In contrast, DRESS is a deterministic, unsupervised framework that produces canonical fingerprints via fixed-point iteration, without any learning or parameter tuning.

2 Generalized-DRESS

Generalized-DRESS is an abstract template for continuous structural refinement. For each edge (u, v) and a symmetric neighbourhood operator $\mathcal{N}(u, v)$:

$$d_{uv}^{(t+1)} = \frac{f\big(\{d_{e'}^{(t)}\}_{e' \in \mathcal{N}(u,v)}\big)}{g^{(t)}(u) \cdot g^{(t)}(v)} \tag{1}$$

where:

- $\mathcal{N}(u, v)$ is a **symmetric neighbourhood operator**—the set of edges aggregated for edge (u, v),
- f is the **aggregation function** over edges in $\mathcal{N}(u, v)$,
- $g(u)$ is the **node norm**.

3 Motif-DRESS

Motif-DRESS is a specialisation of Generalized-DRESS that fixes $f = \mathrm{sum}$ and $g =$ geometric norm, defines $\mathcal{N}(u, v)$ as the set of edges that co-occur with (u, v) in a specific structural motif M, and introduces an optional weight function $w(e)$ to handle weighted graphs:

$$d_{uv}^{(t+1)} = \frac{\displaystyle\sum_{e' \in \mathcal{N}_M(u,v)} w_{e'}\, d_{e'}^{(t)}}{\|u\|^{(t)} \cdot \|v\|^{(t)}} \tag{2}$$

where $\|u\|^{(t)} = \sqrt{\sum_{e' \in \mathcal{N}_M(u,u)} w_{e'}\, d_{e'}^{(t)}}$ is the weighted node norm at time t.

The weights $w(e)$ act as a multiplicative factor, controlling how much structural information flows along each edge. Because the weights appear identically in the numerator and denominator (both are degree-1 in $w \cdot d$), uniformly scaling all weights does not change the fixed point; only the relative weights matter.

Different choices of motif M yield different neighbourhood operators:

- **Triangle ($M = K_3$):** $\mathcal{N}(u, v) = \{(u, x), (x, v) \mid x \in N[u] \cap N[v]\}$—this recovers Original-DRESS.
- **4-cycle ($M = C_4$):** Aggregates over edges participating in 4-cycles with (u, v).
- **K_4 clique:** Aggregates over edges forming a K_4 with (u, v).

3.1 Properties

Complexity: $\mathcal{O}(\text{Motif Extraction}) + \mathcal{O}\big(I \cdot \sum_e |\mathcal{N}_M(e)|\big)$, where I is the number of iterations and $\sum_e |\mathcal{N}_M(e)|$ is the total size of all motif-neighbourhood lists. For Original-DRESS (triangles), this reduces to $\mathcal{O}(I \cdot m \cdot \Delta)$. For motifs such as 4-cycles or K_4 cliques in sparse graphs, the motif-neighbourhood lists can be significantly smaller, making Motif-DRESS faster than Original-DRESS.

Invariant: $d_{uu} = 2$**.** The self-similarity is invariant under iteration: since $\mathcal{N}_M(u, u)$ includes all edges incident to u symmetrically, and $d_{ux} = d_{xu}$, substituting $v = u$ yields $d_{uu}^{(t+1)} = 2g(u)^2/g(u)^2 = 2$ for all t. Thus $d_{uu} = 2$ is a fixed property of the equation, not a free parameter.

Initialisation. Because Motif-DRESS is scale-invariant (degree-0 homogeneous), the fixed point is independent of the initial scale. Any uniform initialisation $d_{uv}^{(0)} = c$ for all edges, with $c > 0$, converges to the same unique fixed point d^*. In practice, $c = 1$ is used.

3.2 Expressiveness

All experiments below use the K_4 clique motif. Since 3-WL provably cannot distinguish SRGs with identical parameters, each successful distinction below demonstrates that Motif-DRESS empirically distinguishes specific graph pairs that 3-WL cannot.

- **Rook vs. Shrikhande:** Successfully distinguishes this pair of SRGs with parameters $(16, 6, 2, 2)$. The Rook graph ($K_4 \square K_4$) contains K_4 cliques while the Shrikhande graph does not, so the K_4-neighbourhood sizes differ per edge.
- **Chang Graphs:** Distinguishes 5 of the 6 pairwise comparisons among the four SRGs with parameters $(28, 12, 6, 4)$: T(8), Chang-1, Chang-2, and Chang-3. The only failure is T(8) vs. Chang-3, which share identical K_4-neighbourhood structure (all edges have the same K_4 count).

3.3 Spectral DRESS: Signal Preservation under Weighted Iteration

Because the Motif-DRESS update is degree-0 homogeneous, uniformly scaling all weights leaves the fixed point unchanged—only the *relative* weight structure matters. Consequently, structurally meaningful edge weights (such as Laplacian eigenvector differences $w(u, v) = 1 + \|\varphi(u) - \varphi(v)\|_2$, where φ collects the top-k eigenvectors of the normalised Laplacian) are not destroyed by the nonlinear iteration but are instead *refined* into the fixed point. This "Spectral DRESS" variant uses the original triangle motif ($M = K_3$), i.e. Original-DRESS with non-uniform edge weights, and empirically distinguishes all six Chang-graph pairs—including T(8) vs. Chang-3, the only pair that the unweighted K_4-motif approach fails on—demonstrating that DRESS acts as a signal-preserving operator: coherent input structure survives the contraction to the fixed point.

3.4 Original-DRESS as a Special Case

The Original-DRESS equation [3] is a special case of Motif-DRESS with the triangle motif ($M = K_3$), $f = \text{sum}$, and $g = \text{geometric norm}$:

$$d_{uv}^{(t+1)} = \frac{\sum_{x \in N[u] \cap N[v]} \left(d_{ux}^{(t)} + d_{xv}^{(t)}\right)}{\|u\|^{(t)} \cdot \|v\|^{(t)}} \tag{3}$$

where $N[u] = N(u) \cup \{u\}$ denotes the closed neighbourhood of u, and $\|u\| = \sqrt{\sum_{x \in N[u]} d_{ux}^{(t)}}$ is the node norm. Note that $d_{uu} = 2$ is invariant under iteration (Section 3.1). This equation converges to a unique fixed point, providing a continuous structural fingerprint for the graph. However, because it aggregates strictly over triangles (common neighbours), it cannot distinguish graphs with identical triangle counts per edge (e.g., Strongly Regular Graphs).

3.5 Proof Sketch of Convergence

Self-loops. Self-loops are added to every node before iteration (i.e., the algorithm uses the closed neighbourhood $N[u] = N(u) \cup \{u\}$). The self-loop edge (u, u) participates in both the aggregation and the node norm; without it, an isolated edge with no common neighbours would produce $g(u) \cdot g(v) = 0$, making the iteration undefined.

Theorem 1 (Convergence of Motif-DRESS)**.** *Let $F : \mathbb{R}_+^{|E|} \to \mathbb{R}_+^{|E|}$ be the Motif-DRESS update operator. If the aggregation function and the node norm satisfy the degree-0 homogeneity condition ($p = 2q$), then F converges to a unique fixed point $d^* \in [0, 2]^{|E|}$ for any non-negative initial state $d^{(0)} > 0$.*

Proof sketch. The argument proceeds in three steps:

1. **Scale Invariance (Degree-0 Homogeneity):** The aggregation function is positively homogeneous of degree p, and the norm of degree q. Since $p = 2q$, the ratio $F(d)$ is homogeneous of degree 0: for any $\lambda > 0$, $F(\lambda d) = F(d)$. Hence the iteration is self-regularizing and cannot diverge or collapse to zero.

2. **Boundedness:** For any valid symmetric neighbourhood operator $\mathcal{N}_M(e)$, the Cauchy–Schwarz inequality (or Hölder's inequality for Minkowski-r variants) bounds the numerator by the denominator. Self-loop inclusion ensures the denominator is strictly positive. Thus $F(d)_{uv} \in [0, 2]$ for all $d > 0$.

3. **Contraction on the Hilbert Projective Metric:** Since F is a positive, degree-0 homogeneous map on the cone $\mathbb{R}_{>0}^{|E|}$, Birkhoff's contraction theorem guarantees that F is a strict contraction under the Hilbert projective metric $d_H(x, y) = \log\big(\max_e x_e/y_e \cdot \max_e y_e/x_e\big)$, provided F maps a bounded part of the cone into a strictly smaller part—which follows from the Cauchy–Schwarz bound above. By the Banach fixed-point theorem on the complete metric space $(\mathbb{R}_{>0}^{|E|}/\sim, d_H)$, the iteration converges to a unique ray, and the boundedness step pins the representative to $d^* \in [0, 2]^{|E|}$.

A complete formal verification of the contraction constant is deferred to the full version; all empirical tests confirm convergence within 20 iterations. □

4 Δ-DRESS

Δ-DRESS breaks symmetry by running standard DRESS on each node-deleted subgraph $G \setminus \{v\}$ for every $v \in V$. The graph fingerprint is the sorted multiset of n converged edge-value vectors (one per deletion), compared by flattening and sorting without any summarisation.

Unlike approaches that modify the DRESS iteration itself (e.g., clamping edge values), Δ-DRESS runs *unmodified* DRESS on structurally altered graphs. Deleting a node from a regular graph produces an irregular subgraph where standard DRESS can now distinguish structure that was hidden by the uniform regularity.

4.1 Connection to the Reconstruction Conjecture

The multiset $\{\{\text{DRESS}(G \setminus \{v\}) : v \in V\}\}$ is directly analogous to the *deck* in the Kelly–Ulam reconstruction conjecture, which posits that graphs with $n \geq 3$ are determined (up to isomorphism) by their multiset of node-deleted subgraphs. Δ-DRESS computes a continuous relaxation of this deck.

4.2 Properties

Complexity: $\mathcal{O}(n \cdot I \cdot m \cdot \Delta)$, where I is the number of iterations per deletion. The computation for each deleted subgraph is entirely independent, making Δ-DRESS embarrassingly parallel.

4.3 Expressiveness

Δ-DRESS achieves expressiveness incomparable to the standard WL hierarchy: it empirically distinguishes specific SRG pairs that confound 3-WL, yet fails on some instances (e.g., co-spectral vertex-transitive graphs) that higher-order methods can separate. We validated this empirically using sorted multiset comparison (no fingerprint summarisation):

- **Rook vs. Shrikhande:** Successfully distinguished ($\text{SRG}(16, 6, 2, 2)$).
- $2 \times C_4$ **vs.** C_8**:** Successfully distinguished (both 2-regular on 8 nodes).
- **Petersen vs. Pentagonal Prism:** Successfully distinguished (both 3-regular on 10 nodes).
- **Chang Graphs:** Distinguished 5 of 6 pairs among T(8) and the three Chang graphs ($\text{SRG}(28, 12, 6, 4)$). The single failure—T(8) vs. Chang-3—is expected, as this pair is not separated even by 2-WL.
- **Paley(9) vs.** $L_2(3)$**:** Failed to distinguish this co-spectral $\text{SRG}(9, 4, 1, 2)$ pair. Both graphs are vertex-transitive, so the multiset of deletion fingerprints is identical—a genuine limitation of the method.

5 Conclusion

The Generalized-DRESS framework provides a powerful, continuous alternative to the discrete Weisfeiler-Lehman hierarchy. Starting from an abstract template (Generalized-DRESS), we showed that Motif-DRESS—a specialisation using structural motifs as neighbourhood operators—converges to a unique fixed point under three sufficient conditions, and that the Original-DRESS equation is recovered as a special case with triangle motifs. Δ-DRESS extends this by running unmodified DRESS on each node-deleted subgraph, connecting the framework to the reconstruction conjecture. Together with Motif-DRESS, Δ-DRESS empirically distinguishes Strongly Regular Graphs that confound 3-WL, while maintaining scalability well below $\mathcal{O}(n^3)$ for sparse graphs. These advancements provide a highly scalable framework for graph isomorphism testing and structural analysis that bypasses the computational bottlenecks of higher-order WL tests. An open-source implementation with bindings for C, C++, Python, Rust, Go, Julia, R, MATLAB, and WebAssembly is publicly available at `https://github.com/velicast/dress-graph`.

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